

Topological Correspondence of Illuminated Sets — I, E, L

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Purpose: provide a compact but rigorous account of the topological relations between the observed illuminated set I (in image space), its enantiomeric mirror E, and the light source set L (in world space). Explain how these relations can be used by MotionPainter's topology module and show concrete integration strategies, telemetry metrics, and experiments.

1. Overview and motivation

MotionPainter operates on temporal frames and synthesizes artifacts (disks[], SDFs, LUTs). Understanding the topology of illumination fields yields robust priors for seeding geometry, improving minimizer stability, detecting symmetry, and guiding hybrid rendering policies. This document explains how to formalize those relations with algebraic topology tools and how to exploit them in practice.

2. Notation and core objects

We use the following notation:

- L — the light source set (a compact subset of $\mathbb{R}^3$, could be an area, surface patch, or collection of patches).
- O — occluder geometry (subset of $\mathbb{R}^3$).
- I — the observed illuminated set on the image plane (subset of $\mathbb{R}^2$) defined by penumbra/umbra classification.
- R — a reflection isometry (plane reflection).
- $E = R(I)$ — the enantiomeric set (mirror image of I).
- $V(p) = \{ s \text{ in } L : \text{segment } p\text{-}s \text{ unobstructed} \}$ the visible-subset of L for image point p.
- $f(p) = \mu(V(p)) / \mu(L)$ the visible-fraction scalar field on I (μ is area/radiance measure).

3. Topological relations: formal statements

3.1 I and E: homeomorphism and induced (co)homology

The reflection R is a homeomorphism $R: \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Therefore R restricts to a homeomorphism between I and E, so I and E are homeomorphic. Consequence: their homology groups and Betti numbers are isomorphic: $H_k(I) \cong H_k(E)$ for all k. Orientation is reversed, so on top-degree cohomology the pullback R^* acts by multiplication by -1 on orientation-sensitive classes.

Practical take: persistence diagrams computed on I and E will match in barcode structure; differences in persistence indicate reconstruction or measurement asymmetry.

3.2 L and I: projection, pushforward, and collapse

The imaging process and occlusion define a (multi)mapping from L to I: rays map points $s \in L$ to image coordinates. Occluders O induce collapses where distinct subsets of L project to overlapping regions in I or are partially hidden. Algebraically, the projection $\pi: L \rightarrow I$ induces a pushforward on homology: $\pi_*: H_k(L) \rightarrow H_k(I)$. Typically dimensions differ (L is 2D/3D, I is 2D), so topological features of L may be collapsed or merged under π_* .

Important categories of behaviour:

- Local homeomorphism: regions where projection is injective (no occlusion) preserve topology locally (π_* is iso on local homology).
- Collapsed cycles: handles or holes in L may map to boundaries or merge in I when occluded.
- Caustics/singularities: fold/cusp catastrophes in projection produce high-curvature features in f and in the boundary of I (these are detectable).

4. Inferring topology of L from I (what is information-preserving?)

Exact inversion of topology is impossible in general without additional constraints (occlusion unknown, limited viewpoints). However, partial inference is feasible under plausible assumptions.

Assumptions that enable inference:

- Known or estimated occluder geometry O (from depth or prime-ends).
- Multi-frame / multi-view observations (camera motion or light motion).
- Priors on L (e.g., L is connected, small genus, or belongs to a parametric family).

Under these, you can recover:

- Number of connected components of L (if distinct bright components map to separated high- f regions across views).
- Boundary features (edge count, curvature concentration) via high-gradient loci in f and their persistence across frames.
- Candidate cycles/handles: persistent holes in the support of f across multi-view reconstructions suggest corresponding features in L (subject to occluder ambiguity).

5. Computational pipeline to extract topological signatures

We propose a practical pipeline that integrates with your existing Topology-aware Prime-Ends module (uploaded to the Vault). The pipeline extracts homology features, persistence diagrams, prime-ends, and a mapping between illumination features and source hypotheses.

Stages:

- 1) Preprocess: obtain calibrated frames, optionally depth, and compute f -map proxies (Monte-Carlo / motion-probing).
- 2) Threshold & multi-scale filtration: build a family of sublevel (or superlevel) sets $I_t = \{ p : f(p) \geq t \}$ for t in $[0,1]$.
- 3) Compute persistent homology (0- and 1- dimensional) over I_t (use Gudhi / Ripser). Record birth/death pairs and persistence.
- 4) Prime-ends coupling: use prime-end decomposition on high-gradient boundary regions to identify approach classes and anchor points (see your Prime-Ends doc).
- 5) Multi-frame linking: match homology features across frames (IoU, Hausdorff, or prime-end correspondences) to build temporal persistence.
- 6) Hypothesize L features: where persistent homology suggests stable cycles/components, generate candidate L topologies constrained by camera geometry and occluder estimates.
- 7) Validate with forward model: simulate projection of candidate L into I (with occluder models) and compare persistence signatures; score hypotheses.

6. Integration points inside MotionPainter

Where to plug this pipeline into Motion[■]Painter:

- PreprocessorWorker: emit fMap proxies, calibrated frames, and lightTrack artifacts.
- Topology Engine (new / existing): consume fMaps & edges; compute persistence diagrams and prime-ends (use uploaded doc workflows).
- Hypothesis Engine: assemble candidate L topologies and produce 3D hypotheses (store in Vault).
- PackingSDF & Minimizer: use homology-informed density maps to seed primitives, preserving important loops and boundaries.
- CompositeRenderer: use topology-aware masks to select blend policies; avoid aggressive LUT edits across persistent homology boundaries.
- Telemetry subsystem: record persistence features as compact telemetry artifacts for automatic policy tuning.

7. Benefits to Motion[■]Painter (practical and measured)

Tracking topological correspondence yields measurable benefits:

- Stability & Perceptual Fidelity: persistent homology identifies illumination features that must be preserved to avoid perceptual artifacts (soft edges, loops).
- Reduced search/optimization space: symmetry priors and known homology reduce degrees of freedom in Minimizer (fewer local minima).
- Robust LUT and slat generation: ensure LUT seam placement respects homology boundaries to prevent tearing or inconsistent lighting across slats.
- Error detection & QA: mismatch between I and its enantiomer E signals reconstruction bias, depth errors, or asymmetric illumination — trigger re-calibration.
- Data augmentation & learning: persistent features provide compact labels for supervised learning of slat placement or frame selection policies.

Quantify benefits with metrics: reduction in disk jitter score, increased PSNR/SSIM on stabilized render, decrease in perceptual flicker (measured via temporal SSIM), and improved optimizer convergence rates.

8. Example experiments and diagnostics

Experiment A: Symmetry test

- Collect short video with a symmetric object under a single extended lamp.
- Compute I and E; compute persistence diagrams for both.
- Compare diagrams: ideally match. Significant differences indicate asymmetric illumination or occluder-induced depth bias.

Experiment B: Recovering a two-patch L

- Simulate or record scene where L is two separated patches (two bulbs).
- Using multi-view recordings, compute per-view fMaps, accumulate projected supports, and compute persistent components. If two persistent components appear across views, hypothesize L has two components.

Experiment C: Prime-ends to handles

- Use prime-ends from uploaded Topology-aware doc to localize high-multiplicity boundary approach classes; where multiplicities persist across frames, propose handles in L and validate with forward simulation.

9. Message contracts, artifacts, and telemetry

Suggested artifacts (additional to previous companion):

- mp:artifact:homology:{frameKey}:v1 — {betti:[b0,b1], persistence0, persistence1, diag0, diag1 (summaries)}
- mp:artifact:primeends:{frameKey}:v1 — prime-end list (ids, anchors, pixel masks, persistence).
- mp:artifact:L_hypotheses:{rangeKey}:v1 — candidate L meshes or parametric descriptions + score.

Telemetry entries:

- persistent_cycles_count, mean_persistence, symmetry_mismatch_score (between I and E), homology_stability_index (over N frames).

10. Algorithms & pseudocode (compact)

```
# Build filtration and compute persistence # Inputs: fMap (float map on image grid),
thresholds T = linspace(0,1,numLevels) for t in T: I_t = { p : fMap[p] >= t } # binary mask
add I_t to cubical complex filtration # compute persistent homology (0- and 1-dim) via
riper/gudhi diagram = persistent_homology(filtration) # extract significant features
(persistence > tau) significant_cycles = [c for c in diagram if c.persistence > tau]
```

Pseudocode to propose L topology from temporal persistence:

```
# Inputs: per-frame significant cycles and prime-ends over frames # Step: cluster persistent
features across frames clusters = cluster_features_over_time(features, iou_threshold=0.4) for
cluster in clusters: if cluster.persistence_mean > persistThresh and cluster.frames_covered >
minFrames: propose_L_topology(cluster) # initialize parametric mesh/hypothesis constrained by
cluster anchors score = forward_simulate_and_score(hypothesis, observed_fmaps) if score >
acceptThresh: persist_hypothesis(hypothesis)
```

11. Implementation notes and libraries

Recommended libraries/tools:

- Homology & persistence: Gudhi, Ripser.py
- Prime-ends, graph routines: use your existing Topology-aware Prime-Ends module (see Vault).
- Optical flow & tracking: RAFT or OpenCV Pyramidal LK for production/real-time respectively.
- 3D reconstruction and mesh handling: Open3D, OpenMVG/OpenMVS.

Engineering tips:

- Run persistence on downsampled/blurred fMaps for speed; refine only on ROIs.
- Store compact persistence summaries instead of full diagrams for telemetry bandwidth savings.
- Use temporal smoothing on fMaps before building filtrations to suppress sensor noise.

12. Risks, failure modes & mitigations

Risks:

- Occluder ambiguity: occlusion can make distinct L features appear merged — mitigated by multi-view or active probing.
- Noise-sensitivity: small fluctuations in f can produce spurious low-persistence cycles — mitigate via persistence thresholding and temporal smoothing.
- Computational cost: persistence on large grids can be expensive — mitigate via multiresolution, ROI processing, and caching.

Failure detection heuristics:

- High symmetry_mismatch_score → trigger recalibration or request additional viewpoints.
- Rapid churn in persistent cycles → lower sampling rate and increase smoothing; indicate insufficient SNR.

13. Suggested rollout plan

Phase 1 (quick): integrate per-frame persistence summary (downsampled fMap) in Topology Engine. Emit mp:artifact:homology.

Phase 2 (mid): link persistence features across time and compute symmetry_mismatch_score; expose UI overlay for debugging.

Phase 3 (long): implement L_hypotheses generation and forward-simulation validation; integrate with Minimizer and AmbiAdapter for topology-aware seeding.

14. References & further reading

This document builds on the Motion■Painter Topology-aware Prime-Ends design in the Vault. See:

- Topology-aware Prime-ends: Algorithm, Video-buffer Extension & Read_write System. [filecite]

Further background (classical): persistence homology texts, catastrophe theory for caustics, and classic works on shadow geometry (caustics and occlusion).

Prepared as a companion to Motion■Painter's canonical documents. If you want runnable notebooks or JS worker stubs that compute downsampled persistence summaries from fMaps, I can produce them as separate artifacts.